

Textured Stimuli Comfort and Response in SSVEP-Based Brain Computer Interface

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Abstract— State of the art steady-state visual evoked potential (SSVEP) brain computer interface (BCI) stimuli are commonly high-contrast, solid color flashing objects which can contribute to visual discomfort and fatigue. The use of low-contrast, textured flashing stimuli is proposed as a more comfortable alternative stimulus presentation paradigm. Eight participants (aged 19-35) were presented with four textured stimuli at varying frequencies, alongside standard stimuli. Results indicate significant effects of stimulus type as well as an interaction between frequency and channel subset on signal-to-noise ratio (SNR) values. Comfort scores consistently favored textured stimuli over high-contrast options at all frequencies. The observed lack of SNR differences between stimulus conditions supports the feasibility of using textured stimuli in BCIs. This study lays a foundation for developing comfortable and effective BCI systems. The promising results of textured stimuli suggest a potential alternative for SSVEP-based BCI systems, emphasizing the importance of balancing neural responses and user comfort in stimulus design.

Keywords— *Brain-computer interface, steady-state visual evoked potentials, brain-machine interface, BCI, BMI, SSVEP, pediatrics, texture, visual processing*

I. INTRODUCTION

Brain computer interfaces (BCIs) translate changes in neural electrical activity recorded by electroencephalography (EEG) into commands that control an external device [1]. Steady-state visual evoked potential (SSVEPs), one control scheme for BCIs, manipulate neural activity by flashing stimuli at specific, constant frequencies [2]. The flashing frequency of the stimulus manifests as synchronized electrical activity in the individual's EEG [2].

It is well established that solid-color, high-contrast stimuli modulate activity in the primary visual cortex, producing reliable SSVEP signals for BCI system use [3]. It has been observed, however, that these high-contrast, solid-color stimuli contribute to visual fatigue in some users and may cause physical discomforts such as drowsiness and headache [4-6].

By using lower contrast, textured stimuli to alleviate such discomforts, this work explores their potential for use in SSVEP-based BCI control. It is hypothesized that the loss in response in the primary visual cortex due to the low contrast of the textured stimuli will be compensated for by additional responses in higher cortical areas, especially the temporal lobe, producing an adequately strong VEP for BCI use [7-8].

II. METHODS

This experiment evaluated a set of four low-contrast textured stimuli for use in SSVEP-based BCI systems in $n=8$ participants, aged 19-35 (mean = 24 ± 6.67). Individual stimuli were presented at three frequencies to determine the comfort and efficacy of textured stimuli as compared to solid-color high-contrast stimuli across two one-hour long stimulus presentation sessions.

A. Stimulus Parameters

A logarithmic set of contrast ratios was chosen to span from a minimum (2:1) to a maximum (15:1) to investigate differences that contrast may play in comfort and signal elucidation based on processing in the visual cortex [9].

In addition to solid color and texture stimuli, two standard pattern reversal stimuli were included to act as well-studied "higher-level" stimuli due to their ability to recruit edge detection mechanisms in the visual processing system [10-11]

Frequency values of 9.6, 16 and 36 Hz were chosen to cover a wide frequency range of VEP stimuli. The low SSVEP frequency (<12 Hz) is represented by 9.6 Hz, the medium SSVEP frequency (12-30 Hz) is represented by 16 Hz. The 36 Hz stimulus represents the high frequency range of SSVEPs [12].

B. EEG Data Acquisition

EEG data was acquired using gel-based scalp electrodes using the g.tec g.GAMMAsys system, g.USBamp amplifier, and g.GAMMAcap headcap. A montage of 16 electrodes (Oz, POz, Pz, Cz, O1, O2, PO7, PO8, P3, P4, P7, P8, T7, T8, C3, C4) were placed in accordance with standard 10-10 system locations with a common reference at Fpz. EEG signals were recorded at a sampling rate of 256Hz.

C. Stimulus Presentation

Grey and green stimuli were tested at four different contrast levels: 15:1, 7.66:1, 3.91:1, 2:1. Each stimulus was presented at flash frequencies of 9.6, 16, 36 Hz. Additionally, six different greyscale stimuli, each presenting a texture or pattern reversal, were tested at three frequency values. Checkerboard and square grating patterns with a spatial frequency of 1 c/deg were presented as a pattern reversal stimuli. The four other texture conditions, 'worms', 'wood

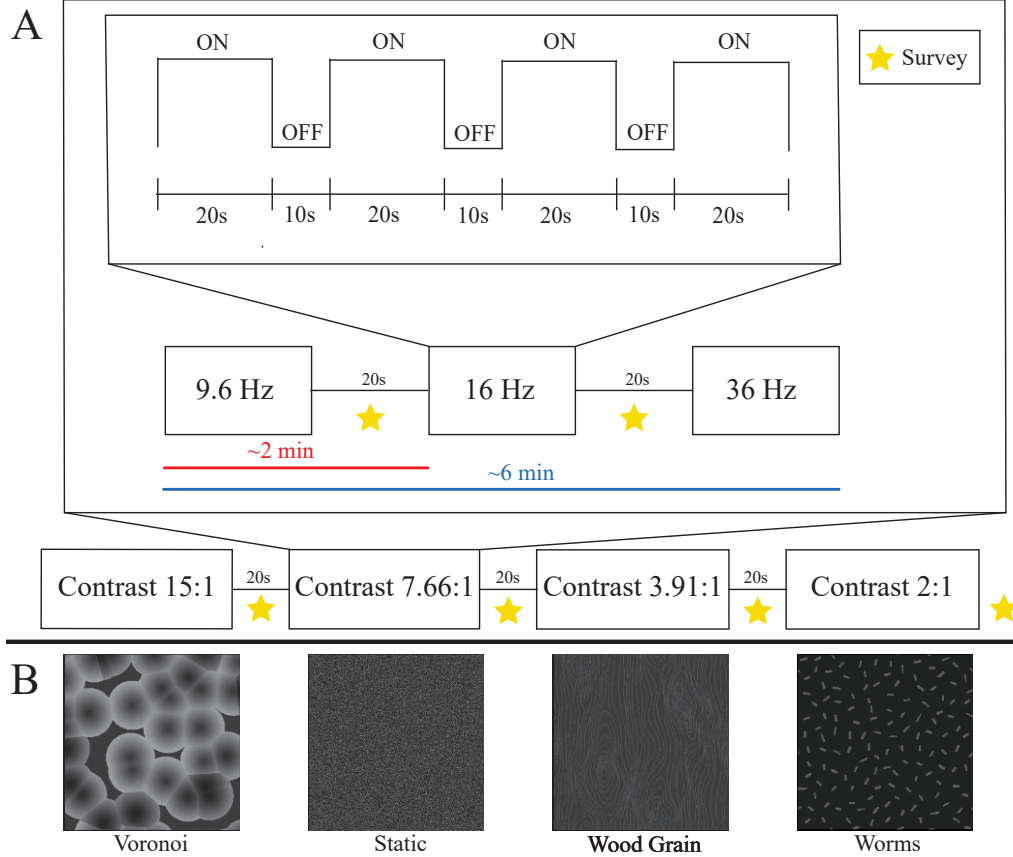


Figure 1. (A) Diagram representing presentation of one complete “set” of solid-color grey stimulus presentation. (B) Textured patterns shown as visual stimuli.

grain', 'static', and 'voronoi' (Figure 1B) were presented as flashing textured stimuli.

Stimuli were presented in the Unity3D Software as an extension of the *BCI-Essentials Unity* available open-source package. In Unity, the above stimuli were presented in “sets” — where a set is designated by presentation of a specific stimulus (e.g., grey stimulus at contrast level 2:1) at each of the specified frequencies (9.6, 16, 36 Hz). All stimuli were presented on a 32” LG monitor with a refresh rate of 144 Hz. The order of stimuli was randomized for each participant. Stimulus presentation proceeded as shown in Figure 1A. Self-reported comfort of each stimulus was reported verbally by the participant during a 20 second break between each trial using a simple 7-point Likert scale that ranged from “Most Comfortable” (value 1) to “Most Uncomfortable” (value 7).

D. EEG Data Processing

EEG data was analyzed using typical best practices with custom python scripts (<https://github.com/kirtonBCIlab/bci-essentials-python>). First, EEG data was visually inspected to remove signals from electrodes which had poor impedance or a significant amount of noise. The data was bandpass filtered between 0.5 and 40 Hz, followed by an independent

component analysis (ICA) to identify and remove spurious or unwanted features causing noise in the signal [13]. To obtain the frequency content of the EEG data, the broadband power spectral density (PSD) of the data was calculated (visualized in Figure 2). The PSD values were used to calculate the signal-to-noise ratio (SNR) of each desired frequency peak in the power spectrum of the EEG data [14]. Prominent harmonics of each of the tested frequencies were included in this calculation, including the 2 harmonics above and below each of the tested frequencies. An SNR value was calculated for each unique stimulus.

E. Statistical Analysis

Prior to statistical analysis, we excluded several conditions of the experiment to proceed with a reduced scope in our statistical analysis. This reduced scope analysis was conducted to be in line with what a future, complexity reduced experiment would look like when extending this work to include children and individuals with atypical neurodevelopment. Stimuli were divided into three groups: maximum contrast, minimum contrast, and texture. The maximum and minimum contrast were the scores for the solid color grey stimuli. The texture scores were the mean of the four textured stimuli.

To investigate the effects of stimulus type and frequency on comfort scores, a 2-way repeated measures ANOVA was used. A 3-way repeated measured ANOVA was used to investigate the combined effects of stimulus type, frequency, and channel subset on SNR. For both tests, the normality of residuals was assessed using the Shapiro-Wilk test prior to analysis. In cases where the assumption of sphericity was violated, the Greenhouse-Geisser correction was applied. The Bonferroni correction was used to determine the significance of pairwise comparisons.

III. RESULTS

A. Signal-to-Noise Ratio (SNR) Analysis

Four different subsets of channels (O, T, OT, All) based on typical EEG neuroimaging regions were identified to determine if the inclusion of temporal lobe channels contributed to the signal created by SSVEP stimulation. The O (occipital) subset includes channels O1, O2, and Oz and the T (temporal) subset includes channels T7 and T8. The OT subset included all channels from the O and T subsets.

For the stimulus and frequency effects as well as their interaction, Mauchly's test did not indicate any violations of sphericity. For the channel subset effect, Mauchly's test, $X^2(5) = 43.70$, $p < .001$, indicated a violation of sphericity. For any interactions that included the channel subset effect, Mauchly's tests indicated a violation of sphericity.

A main effect of stimulus type on SNR value, $F(2,14) = 5.14$, $p = .021$, was observed. No differences were observed between stimulus types, however. There was an interaction between frequency and channel subset $F(2.52, 17.65) = 7.23$, $p = .003$. The differences in SNR value between channel subsets were investigated at each frequency.

At 9.6 Hz, the O channel subset had an SNR value 1.37, 0.73, 1.71 dB higher than the All, OT, and T channel subsets, respectively ($p = .013$, $p = .024$, $p = .021$). The OT channel subset had an SNR value 0.65, .098 dB higher than the All and T channel subsets, respectively ($p = .049$, $p = .020$). At 16 Hz, the O channel subset had an SNR value 0.92, 2.29 dB higher than the OT and T channel subsets, respectively ($p = .019$, $p = .021$). The OT channel subset had an SNR value 1.37 dB higher than the T channel subset ($p = .022$). At 36 Hz, there were no differences between the SNR values of any pairs of channel subsets.

B. Comfort Score Analysis

For the stimulus effect, Mauchly's test, $X^2(2) = 9.63$, $p = .008$, indicated a violation of sphericity. For the frequency effect and the interaction between stimulus type and frequency, Mauchly's test did not indicate a violation of sphericity.

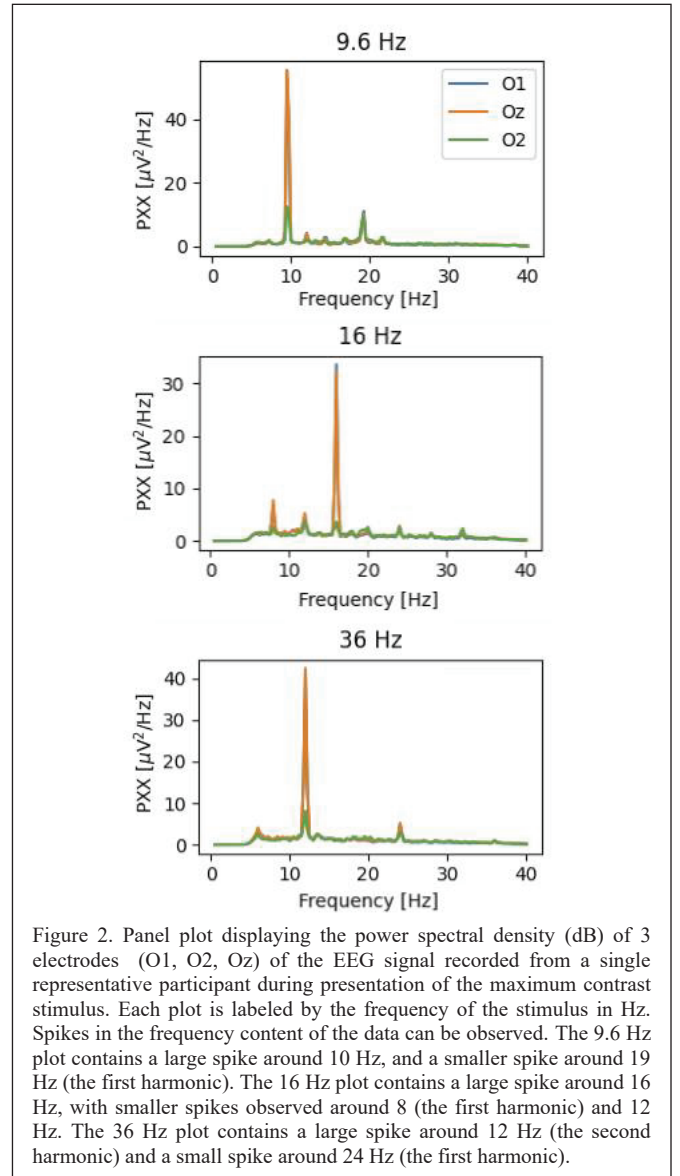


Figure 2. Panel plot displaying the power spectral density (dB) of 3 electrodes (O1, O2, Oz) of the EEG signal recorded from a single representative participant during presentation of the maximum contrast stimulus. Each plot is labeled by the frequency of the stimulus in Hz. Spikes in the frequency content of the data can be observed. The 9.6 Hz plot contains a large spike around 10 Hz, and a smaller spike around 19 Hz (the first harmonic). The 16 Hz plot contains a large spike around 16 Hz, with smaller spikes observed around 8 (the first harmonic) and 12 Hz. The 36 Hz plot contains a large spike around 12 Hz (the second harmonic) and a small spike around 24 Hz (the first harmonic).

There was a main effect of stimulus type on the comfort score, $F(1.11, 7.78) = 18.20$, $p = .003$. It was determined that the maximum contrast stimulus was reported to be significantly less comfortable than the minimum contrast having a comfort score 1.71 points higher on average ($p = .039$). The maximum contrast stimulus was reported to be significantly less comfortable than the textured stimuli having a comfort score 2.57 points higher on average ($p = .005$). The minimum contrast stimulus was reported to be significantly less comfortable than the textured stimuli having a comfort score .87 points higher on average ($p = .001$). There was no effect of frequency and no interaction between frequency and stimulus type was observed.

IV. DISCUSSION

Although a main effect of stimulus on SNR was observed, there were no significant pairwise differences between the stimulus types. It is possible that the sample size ($n = 8$) may

not be large enough to detect differences between individual stimulus types. There was no interaction between stimulus type and channel subset contrary to the hypothesis that the textured stimuli would induce an increased response in the T channel subset. It was found that at 9.6 Hz, the O channel subset had the highest SNR values of all the channel subsets, in line with previous SSVEP literature [4]. At 16 Hz, a similar effect was observed, with the O channel subset having higher SNR values than the OT and T subsets.

Comfort scores demonstrated significant differences between stimulus conditions. Although there was no effect of frequency on comfort score, it was found that the maximum contrast stimulus was rated as the least comfortable stimulus type while the textured stimuli were rated as the most comfortable.

The results suggest that low-contrast, textured stimuli may offer a promising alternative for SSVEP-based BCIs, potentially providing a comfortable and effective interface for users. The observed lack of differences in SNR values between stimulus types indicates that the neural response to textured stimuli may be comparable to traditional high-contrast stimuli, supporting the feasibility of their use in BCI systems.

As the experiment is intended to be extended to a pediatric population, the current results serve as a foundation for designing stimuli that are both comfortable and effective for children. The decision to streamline data analysis was crucial for practical reasons but underscores the need for future comprehensive investigations.

V. LIMITATIONS & FUTURE WORK

The decision to perform statistical analyses on only a subset of the presented stimuli was driven by several considerations. First, as the number of pairs increases, the false discovery rate of paired t-tests increases, and given the variety of stimuli and frequencies tested, an exhaustive analysis was deemed impractical. Second, the primary focus was on addressing the initial research questions regarding the strength of VEPs and the perceived comfort of textured stimuli compared to traditional solid-colored stimuli. Lastly, considering the extension of the experiment to participants aged 5-18 ($n = 35$), the duration of stimulus presentation needed to be shortened to a feasible 30 minutes, necessitating a reduction in the number of stimuli included and analyzed.

VI. CONCLUSION

This study contributes to the ongoing development of BCI technology, emphasizing the importance of considering both neural responses and user comfort in stimulus design. The combination of innovative textured stimuli and careful consideration of channel subsets holds promise for creating

more inclusive and user-friendly BCIs, especially for individuals with motor impairments, including children.

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